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Delegation Decisions in Finance

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Abstract. Based on an online experiment with a sample of finance professionals and participants from the general population (acting as clients), we examine drivers and motives of clients' choices to delegate investment decisions to agents. We find that clients favor delegation to investment algorithms, followed by delegation to finance professionals compensated with an aligned incentive scheme, and lastly to finance professionals receiving a fixed payment for investing on behalf of others. We show that trust in investment algorithms or finance professionals, and clients' propensity to shift blame on others increase the likelihood of delegation, whereas clients' own decision-making quality is associated with a decrease in delegation frequency.

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Keywords: experimental finance • finance professionals • delegation decisions

1. Introduction

Given the complexity of financial products and markets, private investors frequently delegate decisions to finance professionals. This delegation involves decisions about portfolio investments, insurance and pension plans, and seeking advice on various other financial aspects. The economic importance of delegated decision making in finance is indicated by the large and growing market for financial advice and decision making on behalf of clients. As of 2021, the financial planning and advice industry in the United States comprises more than 144,700 businesses and employs more than 230,000 personal financial advisors. The industry revenues have been increasing at an annualized rate of 3.6% over the past five years, to an estimated \$56.9 billion in 2021.¹

When exploring the manifold motives for financial delegation and the relationship between investor characteristics and the demand for financial advice, many studies focus on clients' financial sophistication (see,

e.g., Bhattacharya et al. 2012, Hackethal et al. 2012, Calcagno and Monticone 2015, Kim et al. 2021), but other motives might play an important role too. Among the set of intuitively evident motivations for delegating financial investments, trust in the agent appears to be one of the most obvious (Guiso et al. 2008). Gennaioli et al. (2015) hypothesize that if money managers are trusted, principals have confidence to let money managers take risk on their behalf. They argue that principals (clients) prefer to hire money managers over investing on their own because they are too nervous or anxious to make risky decisions themselves. Another motive for delegating investments is the possibility to blame the agent if the investment should not turn out as expected.² Yet another potential determinant of whether or not to delegate investment decisions is the clients' confidence in their own skills and sophistication, particularly relative to the expected proficiency of the agent. It stands to reason to hypothesize that overconfident clients are less likely to

delegate their investments (see, e.g., Guiso and Jappelli 2006, Bobadilla-Suarez et al. 2017).

Whether individual-level motives and characteristics actually drive the propensity to delegate investment decisions remains an empirical question. In the same vein, the question whether clients' delegation propensity varies with different types of agents and with the incentive schemes faced by the agents arises naturally and calls for empirical investigation. Regarding the latter, misaligned incentives between clients and agents have been portrayed as a driver of excessive risk taking (Rajan 2006, Diamond and Rajan 2009, Dewatripont and Freixas 2012). When this debate has spilled over to the public, trust in the financial industry has suffered (Sapienza and Zingales 2012). This debate might also be one of the reasons why robo advisors have gained relevance in recent years. They promise affordable advice (D'Acunto et al. 2019), although algorithm aversion might hinder their acceptance (Germann and Merkle 2019). Given the economic relevance of delegated decision making in finance, it seems pivotal to better understand the determinants of delegation decisions and their implications. Particularly, we deem it relevant to empirically examine whether clients prefer human money managers—conditional on their incentive structure—over investment algorithms and which personal characteristics drive delegation choices.³

It is our aim to investigate the interplay of clients' delegation decisions—moderated by their preferences, skills, and individual characteristics—and the type of money manager (investment algorithm or finance professionals facing different incentive structures) they can delegate to. We report from a delegation experiment with a sample of Swedish finance professionals and a representative sample of the Swedish general population. We implemented six experimental conditions, varying (i) the pool of participants enrolled (finance professionals or general population) and (ii) the type of money manager the principal can delegate to (investment algorithm, finance professional with aligned incentives, and finance professional with fixed payment). In 25 investment decisions, participants had to allocate an endowment across two or five investment alternatives that differed in their expected payout, riskiness, and diversification potential. Participants from the general population were thereafter given the opportunity to delegate their decisions by replacing their own investments with those of an agent (a finance professional or an investment algorithm, depending on the treatment condition) for determining their payout from the experiment. Invitations were sent out via *Statistiska centralbyrån* (SCB; Statistics Sweden). A total of 408 finance professionals and 550 people from the general population completed the experiment. The collaboration with SCB allowed accessing a stratified sample of finance professionals, restricted to financial analysts, investment advisors,

traders, fund managers, and financial brokers. To the best of our knowledge, we are first to experimentally examine the drivers of delegation decisions in a financial context utilizing a specialized sample of finance professionals—that is, skilled employees that take financial decisions on behalf of clients in their day-to-day work—and a general population sample instead of student participants. Additionally, we obtained a set of predefined variables of the participants' register data for those who completed the experiment from SCB.

Our study provides the following insights. First, we show that clients are most likely to delegate their investment decisions to an investment algorithm, followed by professionals facing aligned incentives and professionals compensated with a fixed payment. However, for clients who decide to delegate, we do not find systematic differences in their willingness to pay attributable to the varying treatment conditions. The latter may suggest that clients who choose to delegate, on average, assign the same value to the virtue of delegation, irrespective of the type of agent. Second, on the aggregate level, we observe that trust is a key determinant for delegating investments. This result is in line with the conjecture of trust being a consistent and major explanatory factor of financial market participation (see, e.g., Guiso et al. 2008). In particular, we find that trust in the agent is explanatory for delegation propensity, irrespective of the type of agent. Moreover, we report that blame shifting motives increase and clients' own decision-making quality decreases delegation propensity. Third, we find that principals tend to ask the agent to take higher levels of risk as compared with the perception of risk they took in their own decisions. This result is in line with the theoretically postulated explanation of finance professionals acting as “money doctors” (Gennaioli et al. 2015). Fourth, we comment on the external validity of our experimental results based on comprehensive robustness analyses. The analysis of principals' self-reported survey responses on real-life delegation decisions to finance professionals and investment algorithms further emphasizes the key role of trust in the agent for delegation decisions. Finally, we discuss potential limitations of our study in the conclusion.

Our study adds to several strands of related research. First, we contribute to the expanding literature on delegation decisions to investment algorithms and robo advice. This emerging literature offers important insights on the impact of robo advice on investors (see, e.g., D'Acunto et al. 2019, Rossi and Utkus 2020b). Previous research, focusing on various fields of applications, provides evidence on algorithm aversion, with people distrusting the advice from algorithms more than advice based on human judgment (see, e.g., Dietvorst et al. 2014, Harvey et al. 2017, Longoni et al. 2019). Yet, evidence on algorithm aversion is mixed.

Germann and Merkle (2019), for instance, find that investors do not have strong preferences whether to tie their incentives to a human fund manager or an investment algorithm; other studies provide evidence on algorithms being preferred over human advice (e.g., Logg et al. 2017). Our findings join the rank of previous indications of algorithm appreciation. We show that participants are more likely to delegate risky decisions to an investment algorithm than to finance professionals in the context of an abstract allocation decision task. In previous studies on the demand of algorithm-based advice, individual-level characteristics play a rather minor role in explaining the heterogeneity in delegation decisions. Instead, clients' financial characteristics (such as, e.g., prior holdings or trading volume) have been more relevant to gauge the reliance on algorithmic advice (see, e.g., Rossi and Utkus 2020b). One noteworthy exception is the survey evidence in a hypothetical setting provided in Oehler et al. (2022), showing that students with, among others, a higher willingness to take risk are more likely to report using robo advice. Similar to our agenda, Bhattacharya et al. (2012) examine the demand side of unbiased algorithmic brokerage. They find that clients who most need financial advice are least likely to obtain it. We contribute to the literature by not only focusing on delegation to investment algorithms, but also comparing (determinants of) delegation decisions to investment algorithms and (human) finance professionals facing different incentive schemes. We show that trust in investment algorithms and a lack of clients' decision-making quality are important channels of delegation decisions to investment algorithms.⁴

Second, we add to the literature on incentives of money managers. A considerable body of theoretical literature has demonstrated that information asymmetries between clients and advisors allow advisors to act in their own interest—to the detriment of their clients (see, e.g., Bolton et al. 2007, Carlin and Manso 2011, Inderst and Ottaviani 2012b). The agency conflicts in the market for financial advisory services modeled in this literature have been supported by various empirical studies (see, e.g., Bergstresser et al. 2009, Hackethal et al. 2012, Mulainathan et al. 2012). In addition, misaligned incentives have been portrayed as major contributors to the financial crisis in 2008 and as main drivers of excessive financial risk taking in general (Frederick 2005, Rajan 2006, Diamond and Rajan 2009, Bebchuk and Spamann 2010, Financial Crisis Inquiry Commission 2011, Dewatripont and Freixas 2012). Yet another strand of experimental studies suggests that even strong financial incentives hardly interfere with agents' attempt to adhere to their clients' preferences (see, e.g., Rud et al. 2018, Ifcher and Zarghamee 2020, Kling et al. 2022). Because the debate about agency conflicts and misaligned incentives has gained widespread public attention and has negatively affected trust in the finance industry (Sapienza and

Zingales 2012), we hypothesize that clients' knowledge about the incentive scheme faced by the agent might play a role for their decisions whether to delegate their investment decisions. As there is virtually no research on the impact of incentives on customers' demand for financial advice, we contribute to the literature by showing that clients prefer delegating to advisors with aligned incentives rather than advisors compensated with a flat payment.

Finally, we add to the literature on trust in the finance industry (Sapienza and Zingales 2012, Zingales 2015) and related concepts such as blame shifting and risk delegation. As noted in the survey by Rossi and Utkus (2020a), hiring financial (human or robo) advice is not only a decision to maximize returns, but to "satisfy a broader set of needs." For instance, Chang et al. (2016) show that delegation reverses the disposition effect by allowing the investor to blame the money manager, making it easier for the investor to sell losing assets. Moreover, stock market participation and financial development have been shown to be conditional on individuals' trust in general and in the finance sector in particular (Guiso et al. 2004, 2008). Whereas trust has also been shown to be an important determinant of financial advice seeking (Lachance and Tang 2012), the results by Georgarakos and Inderst (2011) suggest that trust in financial advice only matters for stock market participation when clients' perceived financial capabilities are low. With respect to delegated decision making in finance, Gennaioli et al. (2015) argue that trust is an important factor for delegating financial decisions in a theoretical framework, and show that increased risk taking is one of the potential motives of clients to delegate their investments to "money doctors." To the best of our knowledge, we are first to empirically examine Gennaioli and colleagues' theoretical account on delegation being driven (partly) by clients' desire to increase risk taking based on incentivized experiments. Our results second the argument by Bucciol et al. (2019) that trust acts as a substitute for risk tolerance: trust in the agent might compensate for a lack in clients' risk-bearing capacities, such that even individuals who shy away from risk can—through delegation—engage in risky investments.

2. Experimental Design

Please note that there is a companion paper (Stefan et al. 2022) that is based on the same experiment. An earlier working paper version presented the results on various aspects related to the delegated investment decisions in a single manuscript (Holzmeister et al. 2019). During the revision process, we split our contribution into two separate papers to discuss our objectives, our results, and their implications in a more targeted manner. In particular, the companion paper

(Stefan et al. 2022) focuses on a different set of variables to address risk communication in delegating risky investments and whether delegation to finance professionals is beneficial for clients in terms of the risk-return trade-off. Because both papers are based on the same experimental implementation and data, substantial parts of the description of the experimental design and the relevant variables as well as their construction are identical in both papers.

2.1. Allocation Decision Task

The workhorse of our experiment is the allocation decision task as used by Banks et al. (2018). The task consisted of 25 investment decisions, in which participants were asked to allocate an endowment of 100 Swedish krona (SEK) on either two or five assets.⁵ Participants were informed about the assets’ returns per 1 SEK invested, depending on whether a coin toss shows up heads or tails. The returns used in the experiment were adopted from Banks et al. (2018), multiplied by a factor of 1.5, and rounded to one decimal place. The returns for each asset in the 25 investment decisions are depicted in Table 1, and the corresponding opportunity sets are illustrated in Figure B1 in online Appendix B.

The task consists of 10 decisions with two binary assets in a first block, and 15 decisions with five binary assets in a second block. Participants were first presented with the task instructions for the first block. After reading the instructions, participants could only continue once they had correctly answered three comprehension questions.⁶ After the first 10 decisions, participants were informed that five rather than two assets would be available for the remaining 15 decisions. The order of the two blocks was fixed for all participants, but the order of decisions was randomized in each of the two blocks. Figure A1 in online Appendix A shows two screenshots of the main experimental task, that is, the allocation decision task, with two and five assets, respectively.

At the end of the experiment, one of a participant’s own or—in case a client opts for delegating the decisions—one of the agent’s decisions was randomly chosen, and a simulated coin toss determined the participant’s payoff. Returns were paid on top of the endowment, that is, final payments could not fall below 100 SEK.

2.2. Decision-Making Quality Index

Similar to Banks et al. (2018), we determine four measures of decision-making quality (DMQI) based on the

Table 1. Return Distributions of the Available Assets in the 25 Opportunity Sets

Set	Asset A		Asset B		Asset C		Asset D		Asset E	
	Heads	Tails	Heads	Tails	Heads	Tails	Heads	Tails	Heads	Tails
1	0.00	1.20	3.60	0.00						
2	3.60	0.00	0.00	1.80						
3	4.80	0.00	0.00	1.20						
4	2.30	0.00	0.00	4.50						
5	0.00	2.40	2.40	0.00						
6	1.20	0.00	0.00	4.80						
7	0.00	2.30	4.50	0.00						
8	0.00	3.60	1.80	0.00						
9	0.00	2.70	3.00	0.00						
10	1.20	0.00	0.00	3.60						
11	0.30	2.70	0.90	0.90	1.20	0.00	0.60	1.80	0.00	3.60
12	0.80	1.50	2.40	0.00	0.40	2.10	1.80	0.80	0.00	3.00
13	2.30	0.60	0.40	1.50	0.00	2.40	1.50	0.90	3.00	0.00
14	0.50	4.10	1.80	0.00	0.00	5.40	0.90	2.70	0.50	0.50
15	2.70	0.30	3.60	0.00	0.00	1.20	0.90	0.90	1.80	0.60
16	2.00	1.20	3.50	0.40	4.50	0.00	0.00	3.00	1.10	2.30
17	1.40	0.20	0.00	1.80	0.50	1.40	0.80	0.80	1.80	0.00
18	2.70	0.50	3.60	0.00	0.90	1.40	0.00	1.80	1.80	0.90
19	0.00	2.40	2.40	0.00	1.80	0.60	0.60	1.80	1.20	1.20
20	0.00	4.50	3.00	0.00	2.00	0.80	0.40	3.50	1.50	2.30
21	0.00	3.60	2.70	0.90	3.60	0.00	1.50	1.50	0.60	2.70
22	2.40	0.40	1.80	0.80	0.00	2.40	3.60	0.00	0.90	1.80
23	0.30	2.70	1.50	0.60	1.20	1.80	2.40	0.00	0.00	3.60
24	5.40	0.00	2.70	0.90	0.50	0.50	0.00	1.80	4.10	0.50
25	0.50	2.70	1.80	0.00	1.40	0.90	0.90	1.80	0.00	3.60

Notes. This table shows the returns (in SEK) per 1 SEK invested for the different assets in the 25 opportunity sets, depending on whether the coin toss shows up heads or tails. Within the blocks of two assets (sets 1–10) and five assets (sets 11–25), the decision problems were randomized in order.

allocation decision task in our experiment. In the following, we provide an overview over the four measures; details are provided in online Appendix C.

First, violations of first-order stochastic dominance (FOSD; Hadar and Russell 1969) are measured by the difference between the expected return of a portfolio chosen by the participant and the maximum expected return of a feasible portfolio that has the same minimum payoff as the chosen one. This idea can be illustrated using the example of opportunity set 1 as described in Table 1: asset A_1 (B_1) yields a payoff of 1.20 SEK (3.60 SEK) if the coin shows up tails (heads) and 0.00 SEK otherwise. Suppose a participant i chooses portfolio $x_{i,1} = (80, 20)$, that is, the participant allocates 80 SEK of the endowment to asset A_1 and 20 SEK to asset B_1 . If the coin shows up heads, participant i receives a payoff of $20 \cdot 3.60 = 72$ SEK, and if the coin shows up tails, the participant receives $80 \cdot 1.20 = 96$ SEK, thus yielding an expected portfolio return of 84 SEK. Apparently, the portfolio $x_{i,1}$ is first-order stochastically dominated: Choosing the allocation $x'_{i,1} = (60, 40)$ would guarantee the same minimum payoff ($60 \cdot 1.20 = 72$ SEK) but a higher maximum payoff ($40 \cdot 3.60 = 144$ SEK). The expected return of this alternative portfolio $x'_{i,1}$ is 108 SEK, thus exceeding the expected return of portfolio $x_{i,1}$ by 24 SEK. This difference in expected returns constitutes our measure of the FOSD violation. Note that choosing portfolio $x_{i,1}$ rather than $x'_{i,1}$ implies that a sizable fraction of the prospective reward—conditional on the fact that both portfolios result in the same minimum payoff—will be left on the table, namely $24 \text{ SEK} \div 108 \text{ SEK} = 22.2\%$.

Second, violations of the generalized axiom of revealed preferences (GARP) are measured using the money pump index (Echenique et al. 2011). The intuition behind our measure is that a decision maker violating GARP could be exploited as a money pump by an arbitrageur who replicates the chosen portfolios at lower cost and sells them to the decision maker at higher prices. Again, this measure can be illustrated extending the example of participant i choosing portfolio $x_{i,1} = (80, 20)$ in opportunity set 1 as described earlier. Now consider opportunity set 10, which is a mirror version of set 1: asset A (B) yields a payoff of 1.20 SEK (3.60 SEK) if the coin shows up heads (tails) and 0.00 SEK otherwise. Suppose participant i chooses portfolio $x_{i,10} = (90, 10)$. The portfolio choices $x_{i,1} = (80, 20)$ and $x_{i,10} = (90, 10)$ are not only first-order stochastically dominated (as per the argument sketched earlier), but also violate the generalized axiom of revealed preferences. The portfolio $x_{i,10}$ yields a payoff of $90 \cdot 1.20 = 108$ SEK if the coin shows up heads and a payoff of $10 \cdot 3.60 = 36$ SEK if the coin shows up tails. Note that the same portfolio is feasible in opportunity set 1 without spending the entire endowment. Allocating 30 SEK of the endowment into asset A_1 and 30 SEK into asset B_1 would replicate the very same portfolio (with payoffs

$30 \cdot 3.60 = 108$ SEK if the coin shows up heads and $30 \cdot 1.20 = 36$ SEK if it shows up tails). An arbitrageur could thus buy the replicated portfolio in opportunity set 1 yielding an expected return of 72 SEK at a “price” of only 60% of the endowment in opportunity set 1 and sell it to participant i at a price of 100% in opportunity set #10, involving a risk-free profit $0.40 \cdot 72 \text{ SEK} = 29 \text{ SEK}$. Following the same line of reasoning, $x_{i,1}$ can be constructed in opportunity set 10. To obtain a payoff of 72 SEK (heads) and 96 SEK (tails), respectively, in opportunity set 10, one would need to invest 60 SEK of the endowment in asset A_{10} and 26.6 SEK in asset B_{10} . Again, a hostile arbitrageur could buy the replicated portfolio (yielding an expected return of 84 SEK) at a price of 86.6 SEK in opportunity set 10 and sell it to participant i at a price of 100 SEK in opportunity set 1, involving a risk-free reward of 14 SEK. Because participant i 's allocation decisions $x_{i,1}$ and $x_{i,10}$ can be replicated at an expense of less than 100% of the endowment in the respective other opportunity set each, the participant leaves a total of $29 + 14 = 43$ SEK—or 24% of the expected returns of the two portfolio decisions—in prospective rewards to a fictive arbitrageur.

Third, financial competence (FC) is a measure of a participant's ability to understand the task at hand. In our experiment, four opportunity sets were presented in both the two-asset (sets 1, 2, 8, and 10) and five-asset frames (sets 11, 15, 18, and 25). Moreover, two of the four opportunity sets presented in the two-asset and five-asset frames, respectively, were constructed as mirror images of one another, that is, only the payoffs for heads and tails were interchanged. Consequently, two opportunity sets were presented four times each ($1 = 10 = 11 = 15$ and $2 = 8 = 18 = 25$). Assuming that a financially competent investor would understand the opportunity sets and invest consistently, we measure a participant's financial competence for identical opportunity sets as the absolute differences between the expected returns of the chosen portfolios.

Fourth, failure to minimize risk (FMR) is based on the assumption of a risk averse investor.⁷ In our experiment for two opportunity sets (5 and 19 in Table 1), the expected return per 1 SEK invested was the same for all assets k . Consequently, all feasible portfolios in these opportunity sets share the same expected return. Thus, choosing a fully hedged portfolio (second order) dominates all other feasible portfolios in these two opportunity sets. A participant's failure to minimize risk in these opportunity sets is measured as the standard deviation $SD_{i,j}$ of the particular portfolio allocation.

Each measure is averaged across all (relevant) opportunity sets. To construct a composite measure of decision-making quality, we conduct a principal component analysis of the four measures described previously (see Table C1 in online Appendix C). In particular, the first principle component constitutes

our decision-making quality index (DMQI). Table C2 in online Appendix C shows the pairwise correlations between the decision-making quality measures for both the general population sample and the finance professionals sample. Violations of FOSD and GARP and, to a lesser extent, the FC measures are positively correlated for both groups. Whereas the FMR measure is also correlated with the other measures for the general population sample, the correlations for the finance professionals turn out to be insignificant.⁸

2.3. Experimental Treatments

Depending on the subject pool, participants were randomly assigned to one of the treatments listed in Table 2. Common to all treatments, both for finance professionals and for the general population sample, is the 25-item allocation decision task, which was detailed earlier.

After having completed all items of the allocation decision task, participants from the general population (principals) had the opportunity to delegate their decisions to an agent. If principals opted for delegating their decisions, the experimental payoff depended on the agent's rather than their own decisions. The design choice that principals made the investment decision first, but were informed about the opportunity to delegate the investment decisions only afterward, warrants further discussion. Although potential clients who do not want to engage in financial matters at all might have dropped out at the outset of the experiment,⁹ our design involves two advantages. First, the opportunity to delegate without prior decisions could have led to delegation in order to receive an experimental payment without spending any effort. Second, our experimental design allows studying whether delegation decisions depend on the decision-making quality of the participants, because investment choices are observed for all participants, irrespective of their decision whether to delegate.

Depending on the treatment, the principals' had the opportunity to delegate either to an investment algorithm programmed by the experimenters (*GP-ALGO*), a finance professional facing aligned, that is, linear, incentives (*GP-ALIGNED*), or a finance professional receiving a flat payment of 200 SEK for deciding on behalf of one or more clients (*GP-FIXED*). Note that, compared with

the baseline condition *GP-FIXED*, treatment *GP-ALIGNED* modifies the incentive structure of the agent, while holding the type of agent constant. In contrast, treatment *GP-ALGO* modifies the type of agent from a human to an investment algorithm.

If principals chose to delegate, they were asked to specify the risk (on a scale from 1 [no risk] to 4 [maximum return]) they wanted to be taken on their behalf by the agent,¹⁰ as well as their willingness to pay for delegating the investment decisions (between 0 and 50 SEK, in steps of 5 SEK). At the end of the experiment, a price for delegating the decision to the agent (between 0 and 50 SEK) was randomly determined. If a participant's willingness to pay was higher than this random number, the participant's decisions were delegated to the agent at the randomly determined price (i.e., the agent's decisions were payoff relevant for the principal); if not, the principal's own decisions were relevant for the payment in the experiment.

Finance professionals were randomly assigned to one of three treatments in which they either made decisions on their own account (*FP-OWN*), or on behalf of (one or more) participant(s) from the general population sample. When deciding on principals' account, finance professionals either faced aligned incentives (i.e., they received exactly the same monetary payoff as (one of) their client(s); *FP-ALIGNED*), or were paid a flat fee of 200 SEK (*FP-FIXED*). Moreover, when deciding on behalf of others, finance professionals were asked to comply with a randomly assigned risk level (between 1 [no risk] and 4 [maximum return]). If participants from the general population delegated their decisions, they were matched with a participant from the finance professional sample, based on the particular treatment and the stated risk level. All details about the delegation decision itself, the risk levels as a means to communicate the desired riskiness of the allocation decisions, the matching modalities, and the payment procedures were common knowledge.

2.4. Questionnaires

After the allocation decisions (but prior to the choice whether to delegate), all participants were asked to self-assess the overall level of risk taken across the 25 items of the allocation decision task on a scale from 1 to 4, that is, on the same scale as when choosing the

Table 2. Treatment Overview

Finance professionals ... make decisions ...		General population ... can delegate decisions to ...	
Condition		Condition	
<i>FP-OWN</i>	on one's own account	<i>GP-ALGO</i>	investment algorithm
<i>FP-ALIGNED</i>	for third party (linear incentives)	<i>GP-ALIGNED</i>	finance professional (linear incentives)
<i>FP-FIXED</i>	for third party (flat payment)	<i>GP-FIXED</i>	finance professional (flat payment)

Notes. This table illustrates the randomly assigned between-subjects treatment conditions for both samples, finance professionals and participants from the general population. The sample sizes per condition are indicated in Figure 1.

risk level when delegating the risky decisions. In addition, we included the following set of nonincentivized survey items at the end of the experiment: all participants were asked about (i) their self-assessed risk attitude in general and in financial decisions (Dohmen et al. 2011, Falk et al. 2016), (ii) their willingness to abstain from something today for a future benefit (Falk et al. 2016), (iii) their trust in mankind in general, in persons from the finance industry, and in investment algorithms, (iv) their proneness to shift blame on others (Wilson et al. 1990), and (v) their level of prosociality in a hypothetical charitable giving setting (Falk et al. 2018). The survey items included in the regression analysis (reported in Table 4) are summarized in Table 3.¹¹

Furthermore, we included a five-item questionnaire on delegation and advice seeking in financial decisions, which was only posed to participants who indicated they have been active in the financial market. Afterward, all participants had four minutes to answer an eight-item Rasch-validated numeracy inventory (Weller et al. 2013), including two questions on cognitive reflection. In addition, participants had to provide their self-assessment of the number of correct answers in the numeracy questionnaire as well as their ranking compared with a random sample of the Swedish population. These assessments allowed us to construct two measures of overconfidence, that is, overestimation and overplacement. Finally, participants had three minutes to answer a six-item financial literacy questionnaire based on van Rooij et al. (2011). For further details regarding the survey items, please refer to online Appendix D.

2.5. Recruitment and Data Collection

We conducted the online experiment in Sweden in cooperation with Statistiska centralbyrån (SCB; Statistics Sweden), which invited participants for the experiment and provided a set of predefined variables of the

participants’ register data for those who completed the experiment. SCB sent out invitations (including a hyperlink to the online experiment and a personalized alphanumeric identifier serving as login credentials) to 8,215 finance professionals and a randomly selected representative sample of 8,215 participants from Sweden’s working general population, excluding finance professionals. The sample of finance professionals consists of financial analysts and investment advisors, traders and fund managers, and financial brokers. For the general population, following Edin and Fredriksson (2000) and Böhm et al. (2018), we only include people with a declared labor income exceeding the minimum amount that qualifies for the earnings-related part of the public pension system. Invitations were sent out in two waves. In the first week of 2019, 20% of the sample were invited. Because no technical issues had arisen, the remaining 80% of the sample were invited in the third week of 2019.

Once participants logged in to the online software, programmed in *oTree* (Chen et al. 2016), using their personal identifier, they were presented with a detailed outline of the experiment. In particular, on the first screen, participants were informed that register data provided by SCB would be matched with the data collected in the experiment. Moreover, participants were informed that the study had been approved by the ethical review boards in Gothenburg and at Statistiska centralbyrån. Participants agreed to the conditions and were directed to the instructions of the experiment. The data handling procedures ensured full pseudonymity of all participants. Further details and additional information on the recruitment, data collection, and experimental implementation are provided in online Appendix A. Complete versions of the experiment and all treatments (in English) are available at <http://hea-2019-01-en.herokuapp.com>.

Table 3. Selected Survey Questions

Variable	Question/statement	Likert scale	
		Min (0)	Max (10)
<i>Risk Tolerance in Financial Matters</i>	I am generally willing to take risks in financial matters.	Does not describe me at all	Describes me perfectly
<i>Trust in Finance Professionals</i>	I generally trust employees from the finance industry.	Does not describe me at all	Describes me perfectly
<i>Trust in Investment Algorithms</i>	I generally trust robo advisors (i.e., computer programs) in financial matters.	Does not describe me at all	Describes me perfectly
<i>Blame Shifting (Others)</i>	If you hurt yourself accidentally, do you sometimes blame somebody who happens to be nearby even though you realize, on reflection, that they were not responsible?	I never blame others	I often blame others
<i>Blame Shifting (Temptation)</i>	Can you easily resist the temptation to blame others for the accidents that happen to you?	I can resist easily	I cannot resist at all

Notes. This table summarizes the Likert items used in the regression analysis below. The table depicts the variable description, the wording of the question/statement, and the corresponding labeling of the minimum and maximum values for each item.

Table 4. Determinants of Delegation Choices

Measure	(1) <i>GP-FIXED</i>	(2) <i>GP-ALIGNED</i>	(3) <i>GP-ALGO</i>	(4) Pooled
<i>Experimental measures</i>				
Decision-Making Quality Index	−0.044 (0.028)	−0.006 (0.026)	−0.078* (0.033)	−0.041* (0.016)
Financial Literacy Score (Std.)	0.027 (0.055)	−0.004 (0.062)	0.003 (0.058)	0.003 (0.034)
Numeracy Score (Std.)	−0.038 (0.062)	−0.030 (0.075)	−0.028 (0.089)	−0.064 (0.044)
Overestimation (Std.)	−0.039 (0.035)	−0.017 (0.035)	−0.047 (0.038)	−0.033 (0.021)
Overplacement (Std.)	0.003 (0.039)	−0.022 (0.042)	−0.015 (0.052)	−0.023 (0.026)
<i>Self-reported measures</i>				
Risk Tolerance (Std.)	0.019 (0.029)	0.011 (0.036)	0.001 (0.036)	0.011 (0.020)
Blame Shifting (Std.)	0.057* (0.028)	0.026 (0.030)	0.053 (0.033)	0.045* (0.018)
Trust in Agent (Std.)	0.068* (0.031)	0.121** (0.033)	0.174** (0.036)	0.123** (0.019)
<i>Controls</i>				
	Yes	Yes	Yes	Yes
Wald χ^2	16.238	18.535	42.016	54.050
$p > \chi^2$	0.236	0.138	0.000	0.000
Pseudo R^2	0.141	0.101	0.218	0.112
Observations	183	185	182	550

Notes. This table reports marginal effects estimates based on logit regressions of the binary choice whether to delegate the investment decision to the agent on a set of experimental and self-reported measures, conditional on the treatment (Models 1–3) and pooled across all treatments (Model 4). Robust standard errors are reported in parentheses. All self-reported measures are standardized scores. “Trust in Agent” refers to a combined variable of trust in finance professionals and financial algorithms, conditional on the treatment. “Blame Shifting” refers to the mean of two standardized survey items on shifting blame on others and resisting the temptation to shift blame on others. “Controls” include gender (binary indicator for female), age (in years), net income from major employment in 2017 (in thousand SEKs), and maximum education level (dichotomous indicators for high school education or less, university education less than or equal to three years, and university education greater than three years). To examine whether the effects systematically differ between treatments, we conduct Wald tests on each covariate after a seemingly unrelated regression with robust standard errors in pairwise comparisons of models. Notably, none of the differences are statistically significant with a p -value smaller than the 0.05 threshold.

* $p < 0.05$; ** $p < 0.005$.

In total, 408 finance professionals and 550 people from the general population completed the experiment. The experiment was conducted in Swedish and took on average 45 minutes to complete. The average payment to participants was 238.9 Swedish krona (SEK; $sd = 122.3$), which is approximately US\$30 given the exchange rate at the beginning of 2019.¹² The experimental data were collected between January 4 and February 10, 2019.¹³

Figure 1 graphically summarizes the sequence of tasks within the experiment. For further information on the main task, please refer to online Appendix B. Details on the side tasks and questionnaires are provided in online Appendix D. Analyses on participants’ decision times conditional on subject pools, treatments, and tasks are summarized in online Appendix F.

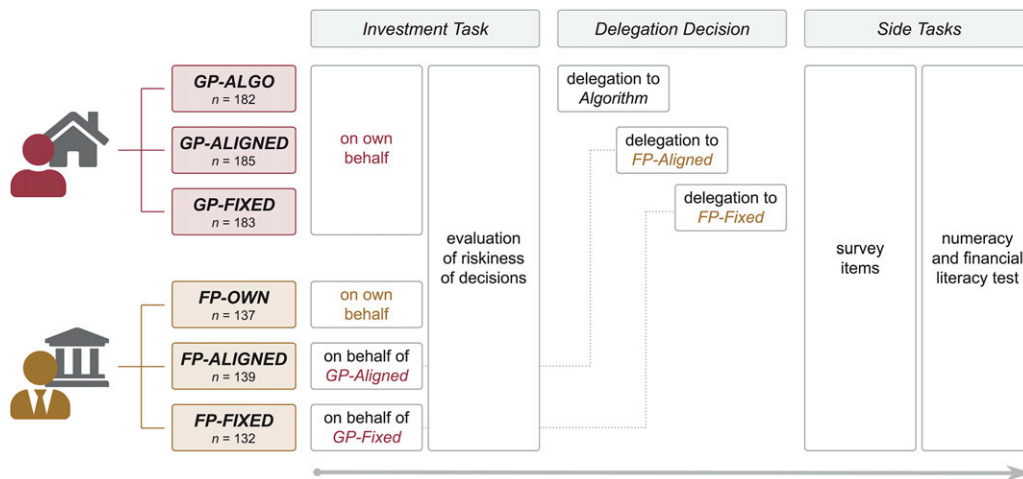
2.6. Register Data

In addition to the data collected in the online experiment, we obtained register data from SCB for each participant who completed the experiment. In particular, we received data on demographics (e.g., age, gender,

income), occupational history (e.g., workplace, firm size), participants’ education, their wealth history, and military records (e.g., scores of the military suitability tests). See online Appendix A for further details on these variables. In the analysis of experimental results, we only use part of the registry data as control variables—in particular, participants’ gender (binary indicator for female), age (in years), net income from major employment in 2017 (in thousand SEKs), and maximum education level (dichotomous indicators for high school education or less, university education less than or equal to three years, and university education longer than three years).¹⁴

3. Results
3.1. Sample Descriptives

Participants in both the finance professional and general population sample are, on average, 42 years old; $sd = 11$ years). The ratios of male participants among finance professionals and participants from the general population are 75.3% and 55.4%, respectively. The

Figure 1. Flowchart of the Experiment

Notes. This figure illustrates the sequence of tasks for participants in our experiment. First, participants were randomly assigned to one treatment and completed 25 investment decisions. Then, participants from the general population could delegate their investment decision to an agent in a delegation decision stage. Finally, all participants completed several side tasks, including self-reported items on economic preferences and supplementary survey questions, a financial literacy test, and a numeracy inventory.

high fraction of males in the finance industry is typical for the job functions under investigation and has also been found in previous studies (see, e.g., Kirchler et al. 2018, Weitzel et al. 2020). The average gross income (from major employment) among finance professionals amounts to SEK 722,046 (sd = 547, 815) and close to 80% hold a university degree. The mean income of the general population is SEK 393,706 (sd = 259,726), with about 50% holding a university degree. On average, participants from the general population took 5.5 and 15.2 minutes for the two-asset and the five-asset decisions in the investment tasks, respectively; the corresponding numbers for finance professionals are 7.0 and 18.4 minutes in treatment *FP-FIXED*, 7.3 and 19.1 minutes in treatment *FP-ALIGNED*, and 5.2 and 15.7 minutes in treatment *FP-OWN*, respectively. Further summary statistics for both samples and details on the times spent on the experimental tasks are presented in online Appendices E and F.

In the following, we first examine principals' decisions to delegate across treatments and then identify potential drivers of delegation decisions. Finally, we discuss the robustness and external validity of our experimental results based on an analysis of participants' self-reported delegation behavior in real-life financial decisions. Note that throughout the presentation of the results, we indicate effect sizes in terms of marginal effects at the means (MEM) and/or odds ratios (ORs) for nonlinear models.

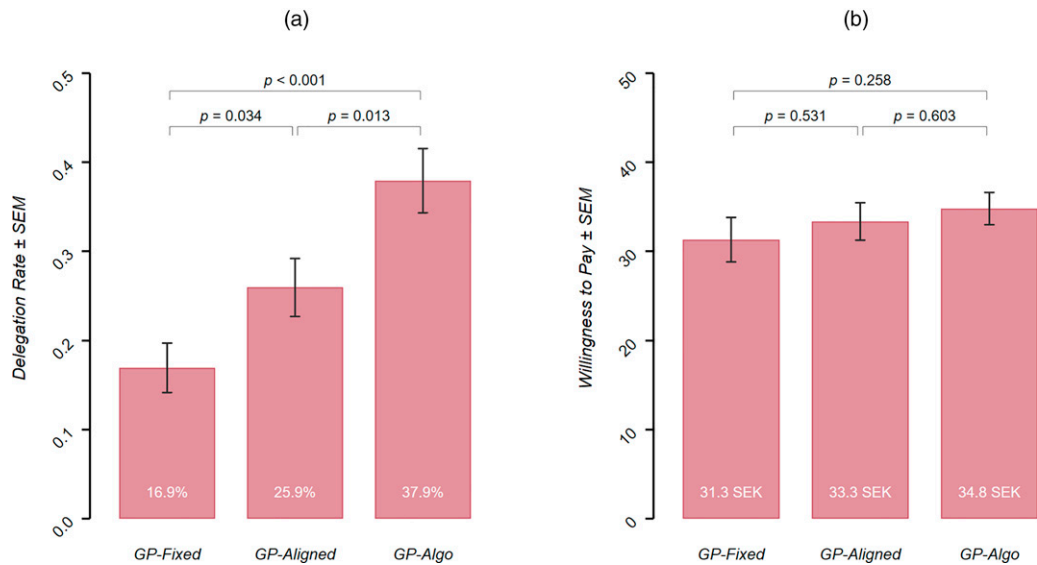
Result 1. *Delegation rates are highest when principals can delegate to the investment algorithm, followed by the treatment in which finance professionals face aligned incentives and the treatment in which professionals are compensated with a flat payment. Principals' willingness to pay for*

delegation (provided they chose to delegate) does neither depend on the type of the agent nor on the agent's incentives.

Support: Figure 2(a) shows the fractions of principals delegating their investment decisions to the agent for each of the three treatments. The estimates are based on logit regressions ($n = 550$; robust standard errors) of the binary delegation choice on treatment indicators. Compared with the delegation rate of 16.9% in treatment *GP-FIXED*, we find that participants are significantly more likely to delegate their investment decision to the agent in treatment *GP-ALIGNED* (25.9%; MEM = 0.090, $p = 0.034$), and in treatment *GP-ALGO* (37.9%; MEM = 0.210, $p < 0.001$). The difference in delegation rates between the treatments *GP-ALIGNED* and *GP-ALGO* is statistically significant too (MEM = 0.120, $p = 0.013$). These results suggest that principals take into account both the type of the agent (i.e., whether the agent is an algorithm or a finance professional) and the agents' incentives (fixed or aligned compensation) in their delegation decisions.

Figure 2(b) depicts principals' mean willingness to pay for delegating their investment decisions to the agent, conditional on the treatment. Note that the willingness to pay is only elicited for participants that chose to delegate ($n = 148$). The estimates are based on an ordinary least squares regression of principals' willingness to pay on treatment indicators (with robust standard errors). Although delegation rates vary considerably across treatments, we report no statistically significant differences between treatments in principals' willingness to pay for delegation. That is, we do neither find evidence that the willingness to

Figure 2. Delegation Decisions Separated by Treatment Conditions



Notes. Panel (a) depicts the share of principals opting for delegating their investment decisions to the agent conditional on the treatment. Error bars indicate standard errors of the mean (SEM); p -values are based on a logit regression of delegation on treatment indicators (with robust standard errors; $n = 550$). Panel (b) shows the mean willingness to pay for delegating the investment decisions of principals who chose to delegate. Error bars indicate standard errors of the mean (SEM); p -values are based on an ordinary least squares regression of willingness to pay on treatment indicators (with robust standard errors, $n = 148$).

pay for delegation (of principals who chose to delegate) differs conditional on whether the agent is a finance professional or an investment algorithm, nor on whether finance professionals face a flat compensation or a linear incentive scheme. Although there is a clear ranking of delegation preferences to agents (i.e., investment algorithms > finance professionals with linear incentives > finance professionals with flat compensation), those clients that actually delegated their decisions might consider delegation as having equal merit, independent of the agents' incentives and whether the agent is an investment algorithm or a human. This might be the reason why the willingness to pay for delegation does not differ across treatments.

Result 1 provides a clear indication that principals' delegation decisions are affected by both the type of the agent and the incentives faced by human agents—that is, factors that have been exogenously varied between the treatments in our experiment. The question of whether the variation across treatments can be explained by the heterogeneity on the participant level arises naturally. Thus, in a next step, we investigate whether the endogenous variability in individual-level characteristics of principals have a systematic effect on the likelihood of delegating financial investment decisions, and whether these effects explain the observed differences in delegation rates between treatments.

Result 2. *Principals' propensity to delegate their investment decisions increases with trust in the agent. On aggregate, blame-shifting motives tend to increase the propensity*

to delegate whereas the principals' decision-making quality tends to decrease the likelihood of delegation. Principals who delegate their investment decisions, on average, request the agent to take more risk than they perceive to have taken in their own decisions.

Support: Table 4 reports the estimates obtained from logit regressions of principals' delegation decisions on various experimental and self-reported measures, conditional on the treatment (Models 1–3) and pooled across all treatments (Model 4). We find that the principals' decision whether to delegate is significantly driven by the principals' trust in the agent.¹⁵ However, we find that the effect size estimates of the trust variable vary considerably across treatments. The odds of delegating one's decision to a finance professional compensated with a fixed payment (GP-FIXED; Model 1) are expected to increase by 75.7% (MEM = 0.068, $p = 0.028$) for a one standard deviation increase in principals' trust in finance professionals. If the agent is a finance professional facing linear incentives (GP-ALIGNED; Model 2), the odds of delegation, on average, increase by 103.3% (MEM = 0.121, $p < 0.001$) for a one standard deviation increase in trust. For the treatment in which clients can delegate their decisions to an investment algorithm (GP-ALGO; Model 3) the effect of trust turns out to be largest: a one standard deviation increase in principal's trust in investment algorithms, on average, gives rise to an increase of 170.5% (MEM = 0.174, $p < 0.001$) in the odds of delegating the investment decision to the agent. Notably, however, the differences in the effect

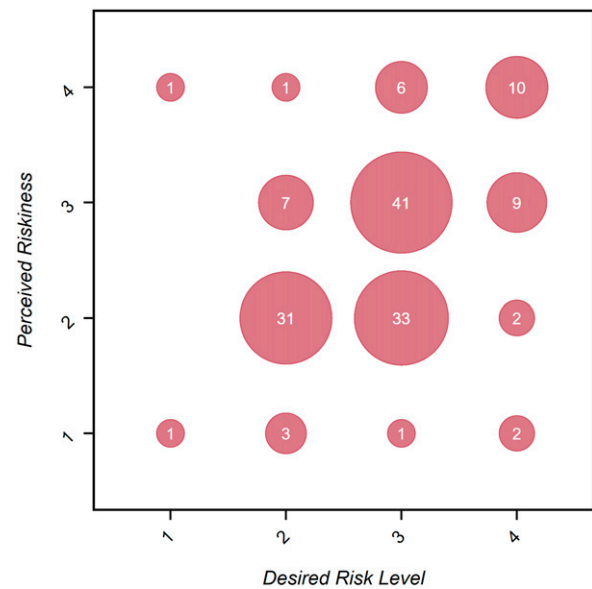
size estimates of trust between treatments—just as for all other coefficient estimates in the multivariate regression analyses—turns out not to be statistically different from zero for all pairwise comparisons (see the table notes of Table 4 for details). In light of these findings, illustrating the relevance of trust for delegating financial decisions, experimental participants' self-reports on trust are particularly revealing. As reported in online Appendix H, trust levels are, on average, higher toward the general population than toward finance professionals—even among their peers—as well as robo advisors.

Apart from trust being a significant driver of delegation decisions, we find that, pooled across treatments (Model 4), blame shifting is significantly and positively associated with delegation propensity, whereas clients' decision-making quality is significantly and negatively related to the likelihood of delegation. On the treatment level, we find that the odds of delegating to a finance professional compensated with a fixed payment (*GP-FIXED*), on average, are expected to increase by 60.5% ($\text{MEM} = 0.057$, $p = 0.041$) for a one standard deviation in blame shifting. Whereas the effect of blame shifting turns out being positive for the other treatments as well, the magnitudes are smaller and the effect is not statistically different from zero (*GP-ALIGNED*: $\text{OR} = 1.163$, $\text{MEM} = 0.026$, $p = 0.389$; *GP-ALGO*: $\text{OR} = 1.360$, $\text{MEM} = 0.053$, $p = 0.108$). With respect to the impact of principals' decision-making quality, we report a significant effect for treatment *GP-ALGO*: for a one standard deviation increase in decision-making quality, the odds of delegating to the investment algorithm are expected to decrease by 36.2% ($\text{MEM} = -0.078$, $p = 0.019$). However, the effect of DMQI on the likelihood of delegation turns out being smaller in magnitude and not statistically significant for the other two treatments (*GP-FIXED*: $\text{OR} = 0.695$, $\text{MEM} = -0.044$, $p = 0.140$; *GP-ALIGNED*: $\text{OR} = 0.963$, $\text{MEM} = -0.006$, $p = 0.803$).

Finally, we report that neither clients' numeracy skills and financial literacy scores, nor the two measures of overconfidence, nor participants' (self-reported) attitudes toward risk in financial matters are statistically significantly associated with principals' delegation decisions—irrespective of the treatment condition.¹⁶

One potential goal behind delegating investment decisions—related to our finding on trust as a driver of delegation—is to increase risk taking. If principals trust the agents, delegation may permit them to take higher risk (Gennaioli et al. 2015). Indeed, our experimental results indicate that principals seek to increase risk taking through delegation. Figure 3 illustrates the principals' desired levels of risk (when delegating their decision to the agent) conditional on the risk perception of their own investment decision. On average, principals tend to ask the agent to take higher levels of

Figure 3. Risk Communication



Note. The figure shows principals' perceived riskiness of their own decisions vs. principals' desired risk levels when delegating their investment decisions to the agent ($n = 148$).

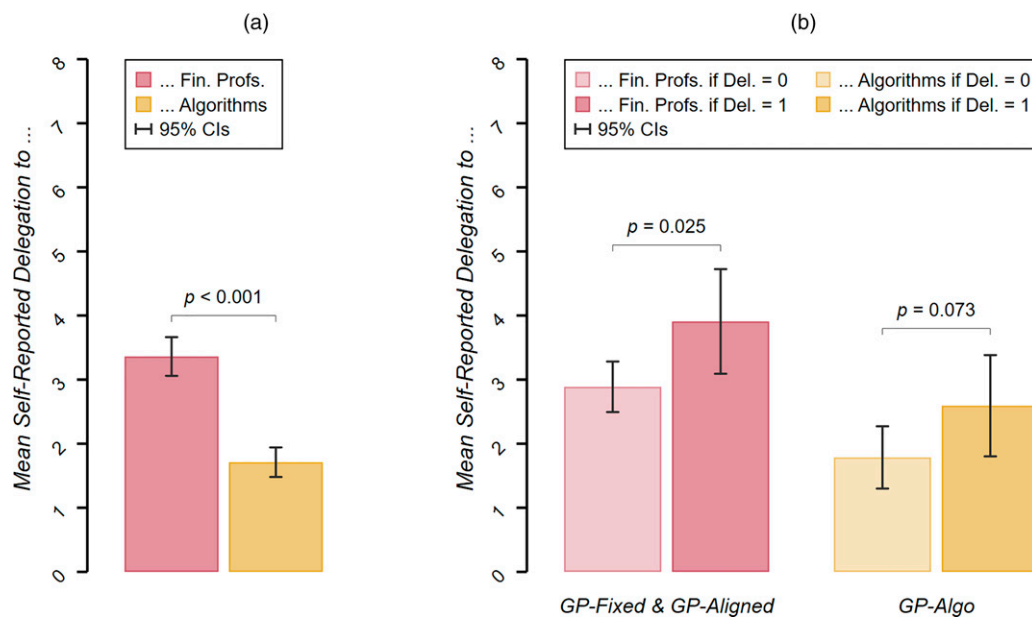
risk ($m = 2.84$, $sd = 0.69$) than they perceive they implemented themselves when deciding on their own behalf ($m = 2.58$, $sd = 0.76$; paired t -test: $t(147) = 4.081$, $p < 0.001$, $n = 148$). This result provides experimental support of the argument by Bucciol et al. (2019) that trust (at least partly) compensates for limited risk-bearing capacities of clients, such that even risk averse individuals—provided that they are sufficiently trusting—might engage in risky investment decisions through delegation.

3.2. Survey Evidence and Discussion

We examine the robustness of the previously described results and provide tentative evidence for their external validity using survey responses on real-life delegation decisions instead of the experimental measures. All Likert scale measures used in the postexperimental survey are described in Table D1 in online Appendix D. In the following, we focus on the two self-reported measures of delegation of investment decisions to professionals and investment algorithms in the general population sample.¹⁷

Figure 4(a) shows the mean responses to the two survey items, emphasizing that the general population sample indicates to delegate more to finance professionals ($m = 3.360$, $sd = 0.154$) as compared with investment algorithms ($m = 1.711$, $sd = 0.118$). The difference in means is statistically significant (paired t -test: $t(404) = 10.494$, $p < 0.001$; $n = 405$). Apparently, this result is in contrast to the results of the main experiment, where participants from the general population

Figure 4. Self-Reported Propensity to Delegate Decisions to Finance Professionals and Investment Algorithms



Notes. Panel (a) shows the mean levels of self-reported real-life delegation decisions to finance professionals and investment algorithms ($n = 405$). The p -value is based on a paired t -test. Panel (b) shows the mean levels of self-reported real-life delegation decisions to finance professionals conditional on the decision whether (Del. = 1) or not (Del. = 0) to delegate the allocation decision in the experiment to a finance professionals in treatments *GP-FIXED* and *GP-ALIGNED* ($n = 267$), and mean self-reported levels of real-life delegation decisions to investment algorithms conditional on the decision whether (Del. = 1) or not (Del. = 0) to delegate the allocation decision in the experiment to the algorithm in treatment *GP-ALGO* ($n = 138$). The p -values are based on two-sample t -tests.

delegate their decisions significantly more often to the investment algorithm (37.9% in treatment *GP-ALGO*) than to finance professionals (16.9% and 25.9% in the treatments *GP-FIXED* and *GP-ALIGNED*, respectively). There are several potential reasons for this difference in revealed algorithm appreciation between the experiment and the survey responses. First, probably few people have been offered to invest using algorithms in real life, as these products are still not too common and usually not strongly promoted by banks.¹⁸ Second, as put forward by our findings and previous contributions to the literature (Gennaioli et al. 2015, Bucciol et al. 2019), trust is an important determinant for delegation decisions, and some participants might put more trust in algorithms constructed by researchers than those designed by financial institutions. Trusting an investment algorithm constructed by a financial intermediary likely calls for an additional “trust premium” to compensate for the frictions (e.g., uncertainty, asymmetric information, technological illiteracy, etc.) that evolve above and beyond trusting a finance professional hired by a financial institution. Third, in the experimental instructions, the algorithm was described to be “programmed in such a way that it maximizes your expected profit conditional on the risk level you indicate [.]” The term “maximizing profit” in the realms of an abstract experimental decision setting could trigger delegation, whereas in real life such a clear promise is beyond reach.¹⁹

Figure 4, however, demonstrates that the delegation decisions in the experiment correlate with self-reported real-life delegation behavior from the survey. In particular, it illustrates the mean survey responses to the two questionnaire items regarding self-reported real-life delegation behavior conditional on the delegation decision in the experiment (“Del. = 0” and “Del. = 1,” respectively), separated by the type of agent participants could delegate to in the experiment (i.e., finance professionals in the treatments *GP-FIXED* and *GP-ALIGNED* and an investment algorithm in treatment *GP-ALGO*). Regarding delegating investment decisions to finance professionals, we observe that the mean survey response of participants choosing to delegate in the experiment ($m = 3.907$, $sd = 0.417$) is significantly higher than the mean response of participants choosing not to delegate ($m = 2.887$, $sd = 0.201$) in the experiment (two-sample t -test: $t(265) = 2.261$, $p = 0.025$; $n = 267$). This difference corresponds to a point-biserial correlation coefficient of $r = 0.138$ between the survey response on delegation to professionals and the observed delegation decision in the experimental treatments *GP-FIXED* and *GP-ALIGNED*. With respect to delegation to investment algorithms, we observe a similar pattern: participants in the *GP-ALGO* treatment who choose to delegate to the algorithm in the experiment tend to report to delegate more to algorithms in real life ($m = 2.593$, $sd = 0.403$) than participants who choose not to delegate ($m = 1.786$,

sd=0.248) in the experiment (two-sample t -test: $t(136) = 1.805, p = 0.073; n = 138$). Although this comparison is not statistically significant—potentially driven by the relatively small sample size of $n = 138$ —the corresponding point-biserial correlation of $r = 0.153$ is comparable to the one on delegation to finance professionals. To sum up, we find that participants reporting to delegate real-life investments to finance professionals (investment algorithms) are more likely to delegate to finance professionals (investment algorithms) in our incentivized experimental task. We consider this as suggestive evidence that the delegation decisions in the experiment involve externally valid patterns.

Finally, we examine individual determinants of self-reported delegation in Table 5. Models (1) and (2) investigate the impact of the set of covariates used in Table 4 on participants' self-reported real-life delegation to finance professionals and investment algorithms, respectively. Just as for the experimental delegation decision, trust turns out to play a key role—in terms of statistical significance as well as effect size—for

delegating investment decisions to both finance professionals and algorithms. With respect to blame-shifting motives, we, again, find qualitatively similar results as for the analysis of the experimental data. In particular, we observe that participants who are more prone to shift blame onto others tend to be more likely to delegate investment decisions to finance professionals (yet, the effect is not statistically significant; $p = 0.065$). As opposed to the effect on delegation decisions observed in the experimental task, we find that both overestimation and risk tolerance significantly decrease participants' self-reported propensity to delegate investments to finance professionals. The negative association of the likelihood of delegation and participants' risk tolerance (in light of the positive effect of trust) integrates well with the argument of Bucciol et al. (2019) that trust (at least partly) compensates for risk aversion: whereas individuals who are willing to bear risk tend to delegate less (and rather make investment decisions on their own), those who shy away from risk appear to be more likely to delegate their investments.

Table 5. Determinants of Self-Reported Delegation Decisions

Measure	(1) Del. to Prof.	(2) Del. to Algo.	(3) Prof. – Algo.
<i>Experimental measures</i>			
Decision-Making Quality Index	−0.166 (0.111)	−0.310** (0.102)	0.129 (0.075)
Financial Literacy Score (Std.)	0.058 (0.264)	−0.070 (0.212)	0.189 (0.254)
Numeracy Score (Std.)	−0.405 (0.367)	−0.130 (0.256)	−0.249 (0.359)
Overestimation (Std.)	−0.446* (0.175)	0.108 (0.139)	−0.552** (0.180)
Overplacement (Std.)	0.226 (0.237)	−0.025 (0.175)	0.261 (0.234)
<i>Self-reported measures</i>			
Risk Tolerance (Std.)	−0.499** (0.164)	−0.081 (0.122)	−0.454** (0.160)
Blame Shifting (Std.)	0.266 (0.144)	0.158 (0.122)	0.123 (0.144)
Trust in Finance Professionals (Std.)	0.955** (0.155)		1.245** (0.172)
Trust in Investment Algorithms (Std.)		0.601** (0.137)	−0.751** (0.177)
<i>Controls</i>	Yes	Yes	Yes
<i>Treatment fixed effects</i>	Yes	Yes	Yes
$F(15, 389)$	8.555	3.272	6.592
$Prob. > F$	0.000	0.000	0.000
Adjusted R^2	0.160	0.086	0.168
Observations	405	405	405

Notes. This table reports estimates from ordinary least squares regressions of the self-reported survey responses whether investment decisions are delegated to (1) finance professionals and (2) investment algorithms on a set of experimental and self-reported measures pooled across all treatments. Model 3 uses the difference between survey responses to the question on delegation to finance professionals and the question on delegation to investment algorithms as the dependent variable, adding a within-subject perspective on how much more likely they are to delegate to finance professionals as compared with algorithms in real life. Robust standard errors are reported in parentheses. All self-reported measures are standardized scores. “Blame Shifting” refers to the mean of two standardized survey items on shifting blame on others and resisting the temptation to shift blame on others. “Controls” include gender (binary indicator for female), age (in years), net income from major employment in 2017 (in thousand SEKs), and maximum education level (dichotomous indicators for high school education or less, university education less than or equal to three years, and university education greater than three years).

* $p < 0.05$; ** $p < 0.005$.

Turning to determinants of real-life delegation decisions to investment algorithms, we find a significantly negative effect of our measure of decision-making quality (DMQI), supporting the result identified in the experimental data.²⁰ As opposed to delegation to finance professionals, delegation decisions to investment algorithms turn out not to be significantly associated with participants' overestimation and risk tolerance. The latter could be interpreted as suggestive evidence that trust only serves as a substitute for risk tolerance in the context of interpersonal relationships (of clients and human agents); in delegation decisions to algorithms, however, trust and risk tolerance may rather be complements (Buccioli et al. 2019), because investors might still feel like investing on their own behalf, even when through an artificial intermediary.

Whereas Models 1 and 2 take a cross-sectional perspective on self-reported delegation decisions in real life, Model 3 in Table 5 employs a within-subject perspective by using the fact that all participants answered both delegation questions. We use the difference between responses to the question on delegation to finance professionals and to the question on delegation to algorithms as an individual-level measure of how much more likely participants are to delegate to a professional than to an algorithm. Again, we find that trust in the respective agent is a main driver: the more (less) participants trust finance professionals (investment algorithms), the more likely they are to delegate investment decisions to finance professionals. It is worthwhile to emphasize the difference in effect sizes: in absolute terms, the (positive) effect of trust in finance professionals on the relative preference for delegating to finance professionals rather than investment algorithms is about 1.7 times as large as the (negative) impact of trust in algorithms. Finally, the difference between delegation frequencies to professionals and algorithms is negatively correlated with overestimation and risk tolerance. Apart from the results on trust, we find that the likelihood to delegate investments to finance professionals rather than to investment algorithms significantly decreases with participants' overestimation and risk tolerance.

Overall, the validity exercise using the self-reported survey responses on real-life delegation decisions to finance professionals and investment algorithms appears to support our main findings based on the incentivized experimental task to a certain extent. In particular, the survey evidence further emphasizes the key role of trust in the agent for delegation decisions.

4. Conclusion

In this paper, we reported from a laboratory-in-the field (online) experiment with finance professionals serving as money managers and participants from the

general population acting as clients. We examined in a controlled experimental environment whether various motivations and individual characteristics discussed in the literature explain clients' propensity to delegate investment decisions to money managers. We showed that clients are most likely to delegate to an investment algorithm, followed by professionals with aligned incentives and professionals compensated with a fixed payment. However, for clients who chose to delegate their investment, we did not find evidence for systematic variation in their willingness to pay for delegation to different types of agents.

We also reported that the variation in clients' decision whether to delegated to the agent can be partly explained by individual-level characteristics. Pooled across conditions, principals' propensity to delegate increased with trust in the agent, no matter whether to a human agent or an investment algorithm. Moreover, we reported that blame-shifting motives increased and own decision-making quality decreased delegation propensity in the full sample. Focusing our attention on the effect of individual characteristics per treatments, blame-shifting motives appeared to be particularly important for clients' delegation propensity to finance professionals with a fixed payment, whereas client's own decision-making quality appeared to primarily affect the delegation propensity to investment algorithm. Finally, we observed that principals tend to ask the agent to take more risk as compared with their risk perception in their own decisions.

Because we chose to study our research questions in a controlled experimental setting, our findings are subject to several limitations. For this reason, we are careful with generalizing our results. Our experimental investment task is an abstraction from real-world investment choices and, thus, differs in several aspects: (i) At first sight, there is no option not to invest the endowment. However, the investment task is designed such that participants could perfectly hedge against all risk, resulting in risk-free allocations. We acknowledge that this implicit outside option might not to be obvious for clients in our setting, which could render delegation to the agent more attractive per se. (ii) Our (experimental) model of the decision environment disregards and (deliberately) bypasses potentially relevant aspects of real-life investment decisions. Likewise, potentially relevant factors for delegation, such as, for example, time constraints, ambiguity aversion, loss-bearing capacities, or inertia, are explicitly ruled out by our design. Whereas the downsides of selection and abstraction apply to basically all experiments and theoretical models in behavioral economics and finance alike, our experimental investigation has several benefits. An abstract investment task with key features of real-world investment situations (e.g., diversification potential, trade-off between risks and returns) offers the upside of

detaching the decision in the experiment from a real-world context, including participants' knowledge about investment products, their beliefs about future developments of markets, and reputation concerns. Moreover, with our design, we can elicit empirically unobservable variables such as the preference for the type of money manager, financial literacy, trust in (human or artificial) agents, and clients' propensity to shift blame on others. Hence, we do not have to rely on proxies, which is a common limitation of observational studies. (iii) We also abstract from introducing tournament components or social comparison—features that finance professionals have been shown to care about (Kirchler et al. 2018, 2020). Furthermore, in our study, we deliberately restrict our attention to aligned versus nonaligned incentives as a first step to introduce information about agents' incentives for clients' delegation decisions as a signal of potential conflicts of interest. We leave the introduction of demand- or supply-side competition and variations and extensions related to the extent of conflicting interests for future research.

Despite these limitations, our study has implications for real-world delegation decisions. Delegating financial decisions is the primary route to financial market participation for many individual investors (see Shum and Faig 2006 for further details). In contrast to theoretical predictions, households' financial market participation is far from universal (Campbell 2006). Thus, if it is a policymaker's goal to promote financial market participation, our results highlight the importance of establishing trust in the finance industry in general and in money managers—including investment algorithms—in particular (Georgarakos and Pasini 2011). Beyond clients' decision-making quality and their propensity to shift blame on others, trust appears to be a consistent and major motive for delegating financial decisions. However, it is not perfectly clear whether clients would actually be better off trusting money managers more: conflicts of interest between clients and money managers (Inderst and Ottaviani 2012a,b), challenges in risk communication, as well as biased and subpar investments by agents (e.g., Kling et al. 2022, Stefan et al. 2022) could constrain the benefits of delegated decision making in the context of financial investments. Thus, the implications of delegated financial decisions are an important area for future research to enrich policymakers' views on the pros and cons of increasing the fraction of delegated investment decisions taken by professional money managers.

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Endnotes

¹ <https://perma.cc/9JXP-N83N>.

² For indirect empirical evidence on blame shifting motives in delegation decisions, refer to Shefrin (2007) and Chang et al. (2016). For a general account on blame shifting, see, e.g., Bartling and Fischbacher (2012).

³ Note that we use the terms “principal” and “client” synonymously throughout the paper for our sample of participants from the general population; likewise, we apply the terms “money manager” and “agent” interchangeably for our sample of finance professionals.

⁴ Note that Bhattacharya et al. (2012) provide evidence that a lack of trust in and/or familiarity with the algorithmic advisor (measured by the length of relationship with the brokerage) and a lack of financial sophistication (proxied by poor past portfolio performance) explain why clients do not seek advice. To the best of our knowledge, we are first to examine individual-level drivers of delegating investments to robo advisors in a controlled experimental environment, which allows to forgo the necessity to rely on proxies of potentially relevant factors in observational studies. Yet, our results on trust and decision-making quality being important factors of delegation decisions to robo advisors appear to integrate well with the findings of Bhattacharya et al. (2012).

⁵ By the end of February 2019, the exchange rate between US dollars and SEK was about 1:9; the exchange rate between the Euro and SEK about 1:10.5.

⁶ Participants who did not answer the comprehension questions correctly, had the opportunity to look at the instructions again until they got the answers right. In addition, they received hints on the correct answers.

⁷ Note that the assumption of risk aversion is explicitly made in this particular measure of decision-making quality. Risk-aversion seems a reasonable assumption to make for our subject pool. In the survey response on financial risk tolerance (see Table 3), participants from the general population provide mean (median) responses of 4.28 (4) on a Likert-scale from 0 to 10 and, thus, can be classified as risk averse on average.

⁸ It could be argued that this might be driven by the fact that the *FMR* measure does not necessarily capture decision mistakes, as failures to minimize risk could also be driven by preferences for risk-seeking. However, we follow Banks et al. (2018) in their choice of components for decision-making quality. Moreover, we use the first principal component of the four measures (in contrast to Banks et al. (2018); see Section C in the Online Appendix) as our *DMQI* variable, with the consequence that uncorrelated influences—such as risk preference—should not pose a systematic issue for our measure.

⁹ For a comprehensive response rate analysis and a discussion of potential self-selection effects, please refer to Online Appendix E.

¹⁰ The investment algorithm was programmed to construct investment portfolios, given the particular risk level, as follows: In each investment decision, the minimum variance portfolio and the maximum return portfolio were mapped to the endpoint options of the risk level scale, i.e., 1 and 4, respectively. Thus, risk level 1 was always associated with a sure payoff, whereas risk level 4 always involved a 100% investment in the asset with the highest expected return. For risk levels 2 and 3, portfolio weights were determined in equally sized steps between these fixed endpoints. For instance, if payoffs were 2.40 SEK/0.00 SEK for asset A and 0.00 SEK/0.80 SEK for asset B, then the risk-free portfolio was characterized by an investment of 25% in A and 75% in B, whereas the maximum return portfolio corresponded to an investment of 100% in A. Risk levels 2 and 3 were associated with portfolios investing 50% and 75% in A, respectively. To the participants the algorithm was described to be “programmed in such a way that it maximizes your expected profit conditional on the risk level you indicate below”. Please note that participants still have to trust that the algorithm operates as described in this statement.

¹¹ The full list of survey items is provided in Table D1 in Online Appendix D.

¹² Thus, the average hourly salary for experimental participants amounts to approximately \$40. This is comparable to other studies with general population samples (e.g., Andersson et al. 2016, 2020). This average annual salary is also comparable—although on the lower end—to other studies with finance professionals (e.g., Haigh and List 2005; Kirchler et al. 2018, 2020; Weitzel et al. 2019).

¹³ In total, only a relatively small fraction of participants—especially for an online experiment—dropped out during the experiment. In sum, 68.9% of all participants that started actually finished the experiment (i.e., 958 out of 1,391). The fraction of completes was 66.3% among the general population and 72.7% among finance professionals, hinting at relatively low and comparable attrition rates across subject pools.

¹⁴ Please note several restrictions in the register data: First, the records for wealth data of SCB end in 2007 and other potentially relevant data such as portfolio holdings of assets or bank account data is not tracked by SCB. Second, adding data from the military suitability tests would have lowered the sample size by close to 40 percent, as all female participants would have been dropped from the analyses. Focusing only on male participants would have lowered the generalizability of our findings for both the general population and the finance industry.

¹⁵ Refer to the table notes of Table 4 for details on the construction of the trust variable.

¹⁶ Note that our finding, that delegation decisions cannot be explained by overconfidence, is in line with the results by Bobadilla-Suarez et al. (2017), arguing that a lack of explanatory power of overconfidence seems to reflect an intrinsic value for choice. Such a “choice premium” may indeed account for part of the unexplained variation in our data as well. Also note that the results by Guiso and Jappelli (2006) indicate that the propensity to delegate relates negatively to risk tolerance. In contrast to these results, we do not find support for a systematic direct effect of risk attitudes on participants’ delegation decisions in our experimental setting.

¹⁷ Note that the two survey questions on delegation were only answered by those participants who did *not* answer the question on frequent Investments with 0 (i.e., “does not describe me at all”). Since 145 participants of the general population sample indicated not to invest frequently at all, the sample size in all analysis presented in this section is $n = 405$ instead of $n = 550$.

¹⁸ As argued, for instance, by Maucijauskaitė (2018), over the course of the last decades the market for robo advice has been on a rapid rise in Sweden—and the European Union in general—due to

considerable technological advances. Yet, the actual use of algorithmic advisory services has been limited so far due to comparably slower behavioral adaption of investors.

¹⁹ For the sake of completeness, we also examine whether the difference between delegation choices in the experiment and the responses to the two survey questions differs between the three treatment conditions. Results are discussed in Online Appendix I.

²⁰ Since *DMQI* is determined based on participants’ choices in the experimental allocation task, the interpretation of the impact of *DMQI* on the answers in the survey questions on delegation is not straightforward. As *DMQI* is constructed as a compound measure of rationality violations in an artificial portfolio context, it might be considered a proxy for financial sophistication. If so, the negative effect of *DMQI* on participants’ self-reports, could be interpreted as suggestive that more financially sophisticated participants are less likely to delegate their investments to algorithms. The interpretation of *DMQI* being a proxy of financial sophistication is supported by significantly positive Pearson correlations between *DMQI* and numeracy ($\rho = 0.137, p = 0.006; n = 405$) and between *DMQI* and financial literacy ($\rho = 0.159, p = 0.001; n = 405$).

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